Python – Pass Keyword, Nested Loops, Multiplication Tables, and Prime Numbers.

3. PASS

3. The pass Keyword in Python

pass is a Python keyword that does nothing when executed. It acts as a placeholder where Python requires a statement, but you do not want to add any action yet.

An indented block cannot be left empty. You can write pass to keep the code valid while you work on it.

if condition: pass while condition: pass for item in items: pass

Note: pass does not skip an iteration or stop a loop. It simply does nothing, and execution continues with the next statement.

In [9]:
```
if (100):pass
print("Pass completed")
```

```
Pass completed
```

In [15]:
```
if(100):
    print("Pass completed")
```

```
Pass completed
```

In [1]:
```
if(1000-1000):pass
```

In [9]:
```
for i in range(1, 6):
    print(i)
    for j in range(1, 4):
        print(j)
    else:
        print("Out of inner")
else:
    print("Out of outer")
```

```
1
1
2
3
Out of inner
2
1
2
3
Out of inner
3
1
2
3
Out of inner
4
1
2
3
Out of inner
5
1
2
3
Out of inner
Out of outer
```

Nested while loop (27:00)

In [15]:
```
i=1
while i<=5:
    print(i)
    j=1
    while j<=3:
        print(j)
        j=j+1
    else:
        i=i+1
else:
    print("Out of outer")
```

```
1
1
2
3
2
1
2
3
3
1
2
3
4
1
2
3
5
1
2
3
Out of outer
```

Inner or Nested Loops

A loop defined inside another loop is called a nested loop. The enclosing loop is called the outer loop, and the loop inside it is called the inner loop.

For each iteration of the outer loop, the inner loop executes until it finishes. A for loop finishes when its iterable is exhausted, while a while loop normally finishes when its condition becomes False.

Python supports four combinations of nested loops:

1. for loop inside a for loop
2. while loop inside a while loop
3. for loop inside a while loop
4. while loop inside a for loop

Indentation identifies which statements belong to each loop.

In [5]:
```python
# Program demonstrating a for loop inside another for loop

for i in range(1, 6):  # Outer loop
    print("-" * 50)
    print("Outer Loop: Value of i = {}".format(i))
    print("-" * 50)

    for j in range(1, 4):  # Inner loop
        print("\tInner Loop: Value of j = {}".format(j))
    else:
        print("-" * 50)
else:
    print("-" * 50)
```

```
--------------------------------------------------
Outer Loop: Value of i = 1
--------------------------------------------------
        Inner Loop: Value of j = 1
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 3
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 2
--------------------------------------------------
        Inner Loop: Value of j = 1
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 3
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 3
--------------------------------------------------
        Inner Loop: Value of j = 1
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 3
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 4
--------------------------------------------------
        Inner Loop: Value of j = 1
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 3
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 5
--------------------------------------------------
        Inner Loop: Value of j = 1
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 3
--------------------------------------------------
--------------------------------------------------
```

In [7]:

```python
# Program demonstrating a while loop inside another while loop

i = 1

while i <= 5:  # Outer loop
    print("-" * 50)
    print("Outer Loop: Value of i = {}".format(i))
    print("-" * 50)

    j = 1
    while j <= 3:  # Inner loop
        print("\tInner Loop: Value of j = {}".format(j))
        j = j + 1
    else:
        print("Inner loop completed.")
        i = i + 1
        print("-" * 50)
else:
```

```
    print("Outer loop completed.")
    print("-" * 50)
```

```
--------------------------------------------------
Outer Loop: Value of i = 1
--------------------------------------------------
        Inner Loop: Value of j = 1
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 3
Inner loop completed.
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 2
--------------------------------------------------
        Inner Loop: Value of j = 1
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 3
Inner loop completed.
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 3
--------------------------------------------------
        Inner Loop: Value of j = 1
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 3
Inner loop completed.
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 4
--------------------------------------------------
        Inner Loop: Value of j = 1
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 3
Inner loop completed.
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 5
--------------------------------------------------
        Inner Loop: Value of j = 1
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 3
Inner loop completed.
--------------------------------------------------
Outer loop completed.
--------------------------------------------------
```

In [9]:
```
# Program demonstrating a for loop inside a while loop

i = 5  # Initialize the outer loop counter

while i >= 1:  # Outer loop: i takes the values 5, 4, 3, 2, 1
    print("-" * 50)  # Print a separator line
    print("Outer Loop: Value of i = {}".format(i))
    print("-" * 50)

    # Inner loop: j takes the values 3, 2, 1
```

```
    # The stop value 0 is excluded; -1 decreases j by 1
    for j in range(3, 0, -1):
        print("\tInner Loop: Value of j = {}".format(j))

    else:  # Runs when the inner loop finishes without a break
        print("Inner loop completed.")
        i = i - 1  # Decrease the outer counter by 1
        print("-" * 50)

else:  # Runs when i >= 1 becomes False, without a break
    print("Outer loop completed.")
    print("-" * 50)
```

```
--------------------------------------------------
Outer Loop: Value of i = 5
--------------------------------------------------
        Inner Loop: Value of j = 3
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 1
Inner loop completed.
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 4
--------------------------------------------------
        Inner Loop: Value of j = 3
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 1
Inner loop completed.
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 3
--------------------------------------------------
        Inner Loop: Value of j = 3
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 1
Inner loop completed.
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 2
--------------------------------------------------
        Inner Loop: Value of j = 3
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 1
Inner loop completed.
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 1
--------------------------------------------------
        Inner Loop: Value of j = 3
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 1
Inner loop completed.
--------------------------------------------------
Outer loop completed.
--------------------------------------------------
```

In [11]:
```python
# Program demonstrating a while loop inside a for loop

# Outer loop: i takes the values 1, 2, 3, 4, 5
for i in range(1, 6):
    print("-" * 50)  # Print a separator line
    print("Outer Loop: Value of i = {}".format(i))
    print("-" * 50)

    j = 3  # Reset the inner loop counter for each outer iteration

    while j >= 1:  # Inner loop: j takes the values 3, 2, 1
        print("\tInner Loop: Value of j = {}".format(j))
        j = j - 1  # Decrease j by 1 after each inner iteration

    else:  # Runs when the inner loop finishes without a break
        print("Inner loop completed.")
        print("-" * 50)

else:  # Runs when the outer loop finishes without a break
    print("Outer loop completed.")
    print("-" * 50)
```

```
--------------------------------------------------
Outer Loop: Value of i = 1
--------------------------------------------------
        Inner Loop: Value of j = 3
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 1
Inner loop completed.
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 2
--------------------------------------------------
        Inner Loop: Value of j = 3
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 1
Inner loop completed.
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 3
--------------------------------------------------
        Inner Loop: Value of j = 3
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 1
Inner loop completed.
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 4
--------------------------------------------------
        Inner Loop: Value of j = 3
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 1
Inner loop completed.
--------------------------------------------------
--------------------------------------------------
Outer Loop: Value of i = 5
--------------------------------------------------
        Inner Loop: Value of j = 3
        Inner Loop: Value of j = 2
        Inner Loop: Value of j = 1
Inner loop completed.
--------------------------------------------------
Outer loop completed.
--------------------------------------------------
```

In [15]:

```python
# Program to generate multiplication tables from 1 to N
# N must be a positive integer

# Read how many multiplication tables the user wants
n = int(input("How many multiplication tables do you want? "))

# Check whether the number is positive
if n <= 0:
    print("{} is invalid input. Enter a positive integer.".format(n))

else:
    # Outer loop: select each number from 1 to n
    for num in range(1, n + 1):
```

```
        print("-" * 50)
        print("Multiplication Table for {}".format(num))
        print("-" * 50)

        # Inner loop: multiply the selected number by 1 through 10
        for i in range(1, 11):
            # Display the number, multiplier, and product
            print("\t{} x {} = {}".format(num, i, num * i))

        else:
            # Runs when the inner loop finishes without a break
            print("-" * 50)
```

```
--------------------------------------------------
Multiplication Table for 1
--------------------------------------------------
        1 x 1 = 1
        1 x 2 = 2
        1 x 3 = 3
        1 x 4 = 4
        1 x 5 = 5
        1 x 6 = 6
        1 x 7 = 7
        1 x 8 = 8
        1 x 9 = 9
        1 x 10 = 10
--------------------------------------------------
--------------------------------------------------
Multiplication Table for 2
--------------------------------------------------
        2 x 1 = 2
        2 x 2 = 4
        2 x 3 = 6
        2 x 4 = 8
        2 x 5 = 10
        2 x 6 = 12
        2 x 7 = 14
        2 x 8 = 16
        2 x 9 = 18
        2 x 10 = 20
--------------------------------------------------
```

In [19]:

```
n = int(input("Enter the Range in which Prime Numbers u want :"))

if n <= 1:
    print("{} is invalid input".format(n))
else:
    print("--------------------------------")
    print("List of Primes within {}".format(n))
    print("--------------------------------")

    for num in range(1, n+1):
        res = True

        for i in range(2, num):
            if num % i == 0:
```

```
                res = False
                break

        if res and num > 1:
            print(num)
```

```
--------------------------------
List of Primes within 4
--------------------------------
2
3
```

In [ ]: